Chapter

Machine Learning-Driven Stacked Ensemble Meta-Learning for Short-Term Solar Power Forecasting

Arif Hussain and Gelli Ravikumar

Abstract

This chapter presents an advanced short-term solar power forecasting method utilizing a machine learning-driven stacked ensemble meta-learning (ML-SEML) framework. The inherent intermittency of solar power poses a significant challenge to accurate forecasting, as traditional machine and deep learning models struggle to capture these fluctuations effectively. To address this issue, we propose a hybrid approach that integrates multiple canonical and ensemble models, including K-Nearest Neighbors (KNN), Multi-Layer Perceptron (MLP), Support Vector Regression (SVR), Gradient Boosting Regressor (GB), and Extreme Gradient Boosting (XGB) within a meta-learning framework. Real-world solar datasets are used to validate the proposed model's performance using key evaluation metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared (R^2) score. Comparative experiments demonstrate that the ML-SEML model consistently outperforms individual learning models and LSTM-based approaches, offering superior resilience, adaptability, and accuracy in forecasting complex solar power patterns. This chapter will provide insights into the design, implementation, and evaluation of ensemble-based forecasting techniques, contributing to enhanced solar energy integration and grid reliability.

Keywords: solar power forecasting, stacked ensemble, meta-learning, prediction, accuracy, model robustness

1. Introduction

Renewable energy (RE) is the need of the day, and clean energy has become the most focused target globally in recent decades. This need and focus of RE in the power market is the limited nature of conventional energy sources such as oil, gas, coal, etc. RE has the advantages of zero fuel costs, low operational and maintenance costs, and low emissions [1]. There are several RE resources like solar, wind, biomass, hydro, etc., among which wind and solar are the most commonly used [2]. The share of solar within the global energy mix reached 5.5%, up from 4.6% in 2022, according to

Ember [3]. Solar energy contributes to reducing global warming and environmental degradation because it emits no greenhouse gases and is also the most suitable RE source with its efficiency, affordability, and abundance.

However, the intermittent nature of solar energy introduces risks and challenges for its utilization as a base load in power systems. The power produced by solar energy systems varies with atmospheric conditions and solar irradiance. Unexpected fluctuations in solar power generation can increase the operational costs of electricity and pose significant risks to the reliability of the electricity supply. Various energy storage systems and complementary generation methods are used to meet the required demand and mitigate the intermittency of solar energy [4]. Integrating solar power into the grid can present several technical issues, such as voltage fluctuations, unintentional islanding, impacts on grid frequency, and power fluctuations within the grid. Several methods exist to address the intermittent nature of solar energy, including various smoothing approaches, control strategies, and accurate solar power forecasting (SPF). The initial and critical solution for managing the intermittency of solar energy is an accurate SPF.

Researchers and industry professionals have developed various SPF solutions to solve these issues; these can be broadly classified as hybrid, data-driven, or physical. These algorithms use a range of input features, such as historical power generation statistics, sun irradiance measurements, and meteorological data, to predict the output of solar PV systems [5]. In the past decade, data-driven machine learning (ML) and deep learning (DL) models have attracted much attention. Ensemble forecasting models are important in ML/DL forecasting tactics because they mix many models and forecast by averaging their results. Averaging forecasts helps reduce server biases when weather data contains outliers, preventing it from making the worst predictions [6]. Multiple methods have been published in the literature employing basic machines and deep learning models, such as KNN, SVR, MLP, GB, LSTM, and CNN [7, 8]. Traditional methods, on the other hand, frequently rely on a single model or basic combinations, making it challenging to capture complicated data patterns and anomalies.

In [9], the author proposes a stacking ensemble method using a recurrent neural network (RNN) as a meta-learner for short-term photovoltaic (PV) power forecasting. This approach combines five models—ANN, DNN, SVR, LSTM, and CNN—and achieves a mean relative error (MRE) of 4.29%, outperforming single models and a benchmark forest ensemble. Similarly, [10] presents a day-ahead PV power forecasting model using residual networks (ResNet) and dense convolutional networks (DenseNet) with meta-learning, though these models are computationally intensive and complex. While methods like stacking ensembles with RNNs or using ResNet and DenseNet models improve accuracy, they come with significant computational complexity due to the need for extensive training and tuning of multiple deep-learning architectures, making them less practical for real-time or large-scale applications.

To overcome this limitation, we present a stacked ensemble meta-learning model for SPF that leverages the capabilities of numerous canonical and ensemble learning models inside a single meta-learning framework. Traditional SPF methods rely on a single model, which can struggle to capture the complex, nonlinear patterns in solar power generation data. By incorporating a diverse set of canonical and ensemble models and utilizing LR as the meta-model, the ML-SEML model offers superior predictive accuracy and robustness. By combining these models into a stacked ensemble architecture, the ML-SEML model can easily manage the irregular and intermittent

nature of solar power data. The meta-model combines the specific characteristics of each base model to generate more accurate estimates. Our proposed framework includes the following contributions:

- We propose a stacked ensemble model (ML-SEML) to enhance prediction reliability by combining multiple base models, each addressing different data characteristics.
- The diverse models in ML-SEML ensure adaptability to varying features and dynamics of solar power generation, providing robustness against fluctuations.
- The proposed model achieves significantly lower MSE and MAE than LSTM and traditional methods, showcasing superior predictive accuracy.
- The proposed model reduces the influence of outliers while being computationally more efficient than LSTM, leading to faster training times and lower resource usage.

2. Methodology

The necessity to capture many facets of the underlying patterns in the data and improve prediction accuracy by using various learning methods motivated the selection of models for our proposed framework. For short-term associations in time-series data, KNN was selected due to its ease of use and efficiency in collecting local data structures. MLP, a fully linked feedforward neural network, is incorporated to simulate intricate non-linear interactions. SVR is appropriate for time-series forecasting with noise since it manages high-dimensional feature spaces well and reduces overfitting. The potent ensemble techniques GB and XGB make high prediction accuracy possible, which combines several weak learners to create strong models. XGB, in particular, is renowned for its effectiveness and capacity to manage intricate patterns and missing data. By integrating different viewpoints from basic and complicated models, we can improve generalization by utilizing the strengths of each model when we include them in a meta-learning framework.

2.1 KNN regressor

A popular instance-based lazy learning classification algorithm, the KNN technique aims to classify test instances according to how closely they resemble a set of k class centers in terms of their characteristics [11]. The key steps in utilizing the KNN algorithm to estimate probabilistic wind speed or power can be summarized as follows:

- Establishing a predetermined gap between the testing and training examples.
- Choosing K nearest neighbors from the training samples with the k lowest distances.
- Using an average approach to predict wind speed or power output.

Euclidean distance, the most widely used method, was employed using the eq. (1):

$$D_{(X,Y)} = \sqrt{\sum_{i=1}^{n} \left(x_i + y_i\right)^2} \tag{1}$$

where X and Y are two instances from the training and testing datasets, respectively; x_i and y_i are input variables, and n is the number of input variables.

2.2 MLP regressor

In 1958, Frank Rosenblatt, the original perceptron creator, built it as a device rather than an algorithm. The perceptron creates a single output that is reliant on numerous actual inputs using a linear combination of input weights [12]. They can be expressed mathematically using eq. (2).

$$y = \varphi\left(\sum_{i=1}^{n} w_i x_i + b\right) \Rightarrow \varphi\left(w^T x + b\right) \tag{2}$$

where w is the weight vector, x is the input vector, b is the bias, and φ is the nonlinear activation function. An MLP's major flaw is overfitting, which is handled by dividing data into training and validation sets. The phases in the perceptron learning process are as follows:

- Measure the total of the inputs after increasing them by a given weight.
- Multiply by the weight, then add a bias factor (0–1).
- Get the perceptron output by feeding the sum via the activation function.

2.3 SVR regressor

SVRs, introduced by Vapnik and colleagues, are unquestionably one of the most popular classification methods [13, 14]. SVR is a supervised method for classification and regression problems that uses hyperplanes to differentiate one data set from another in classification tasks. The weighted sum of training and bi-as outcomes is a class decision function connected with a hyperplane using eq. (3).

$$y(x) = w^T \varphi(x_n) + b \tag{3}$$

where w is the normal hyperplane weight vector, bias is represented by b, and x_n is a true dimensional function vector.

2.4 GB regressor

A powerful ensemble learning technique for regression and classification problems is gradient boosting (GB). GB, which Jerome Friedman developed in 1999,

builds models one after the other, aiming to correct the errors of the previous models [15]. The key benefit of GB is that it can create extremely accurate models due to its durability and versatility. It is possible to balance bias and variance and lower the likelihood of overfitting by carefully regulating the learning rate and the number of iterations. Eq. (4) represents the GB algorithm.

$$F_m(x) = F_{m-1}(x) + \eta h_m(x) \tag{4}$$

where: $F_m(x)$ is the current model, $F_{m-1}(x)$ is the previous model, η is the learning rate, $h_m(x)$ is the new base learner trained on the residual errors.

2.5 XGB regressor

Chen and Carlos improved GB in 2016 to increase performance using innovative optimization methods called Extreme GB [16]. Because of its precision, scalability, and versatility in regression and classification, it has emerged as one of the most used machine learning algorithms. XGB algorithm's key advantages are its capacity to handle complicated data connections, regularization approaches to minimize overfitting, and parallel processing for efficient calculation.

In each iteration, XGB minimizes a regularized objective function for eq. (4) that includes both the loss function and a penalty for model complexity. This objective function is given by:

$$\mathcal{L}(F) = \sum_{i=1}^{n} \ell\left(y_i, \hat{y}_i\right) + \sum_{k=1}^{K} \Omega(h_k) \tag{5}$$

where $\ell(y_i, \hat{y}_i)$ is the loss function (e.g., mean squared error) for the prediction y_i, $\dot{\Omega}(h_k)$ is the regularization term for the k-th tree, which helps control overfitting.

2.6 Proposed ML-SEML model

There are various methods in ensemble learning for utilizing multiple models, including bagging, boosting, and stacking. In our proposed research, we utilize a stacked ensemble where the model combines the predictions of multiple base models using a meta-model, which learns optimally based on the weights of the base model. The definition of meta-learning is "learning to learn," which means that in meta-learning, a meta-model is trained to learn from the output of multiple base models [17]. The strength of individual models makes the stacked ensemble model more robust and accurate for forecasting. In our proposed ML-SEML model, we utilize canonical and ensemble models as base models, which include KNN, MLP, SVR, GB, and XGB, while for the meta-model, we utilize LR. By combining the predictions of these models using the meta-learning LR model, we can improve the accuracy and robustness of SPF. The meta-model uses the base models' predictions as input features to produce the final prediction. This method enables the meta-model to capture intricate interactions between the base models' predictions, perhaps leading to better performance.

Mathematically, let $\{M_1, M_2, \ldots, M_k\}$ represent the base models. The prediction of the i-th base model for an input x is $y_i = M_i(x)$. The meta-model M_{meta} then takes the predictions $\{\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_k\}$ as input and produces the final prediction:

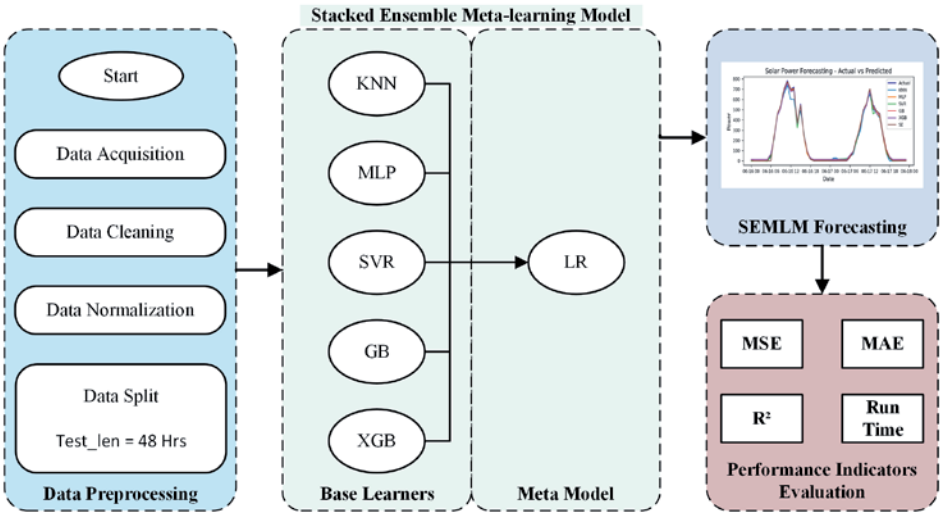


Figure 1.
Flowchart of proposed ML-SEML model.

$$\hat{y} = M_{\text{meta}}\left(\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_k\right) \tag{6}$$

For Linear Regression as the meta-model:

$$\hat{y}_j = \beta_0 + \sum_{i=1}^{k} \beta_i \hat{y}_{i,j} \tag{7}$$

The proposed ML-SEML model is divided into five segments, as illustrated in the flowchart in **Figure 1**. Initially, data preprocessing was carried out, including data acquisition, cleaning, normalization, and data splitting. In the second step, we have a layer of base models, including KNN, MLP, SVR, GB, and XGB. The output of base models is combined using the meta-learner LR model in the third step. In the fourth step, the forecasting is performed based on the ML-SEML model, and finally, the model is evaluated using evaluation metrics.

3. Case study

3.1 Data description

We analyzed data from a solar power plant in India during 34 days [18]. It consists of two file pairs, each including one power generation dataset and one weather sensor reading dataset. Power generation statistics are acquired at the inverter level, with each inverter connected to many lines of solar panels. Sensor data is collected at the plant level, and a single array of sensors is strategically positioned. The data include ambient and module temperature, irradiation, AC and DC power, daily and total yield, and the number of source keys. A correlation is a statistical term that reflects the degree of the link between two variables in terms of positive and

negative correlations. Correlation can be utilized to test theories regarding the causes and effects of relationships between variables, as shown in **Figure 2**. The seasonal decomposition of DC power is also shown in **Figure 3**, in terms of trend, seasonal component, and random residuals.

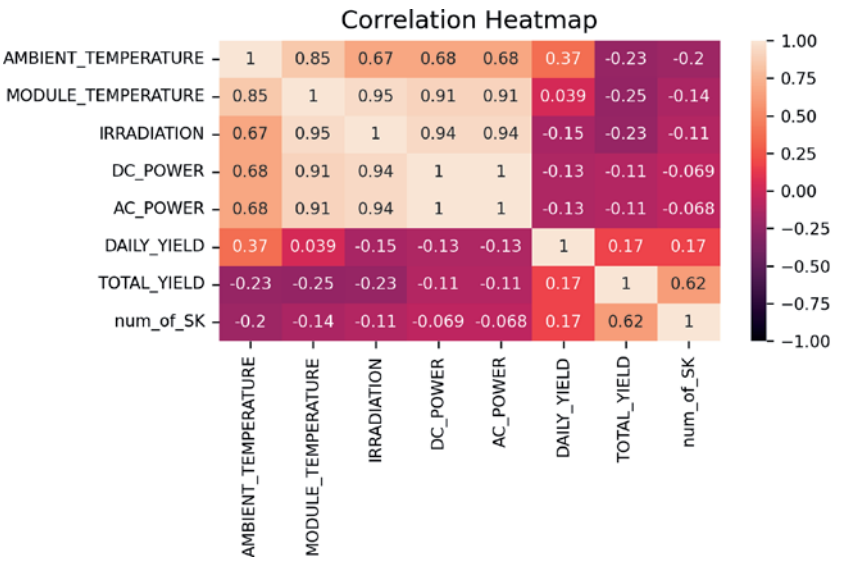


Figure 2.
Correlation matrix between variables of solar data.

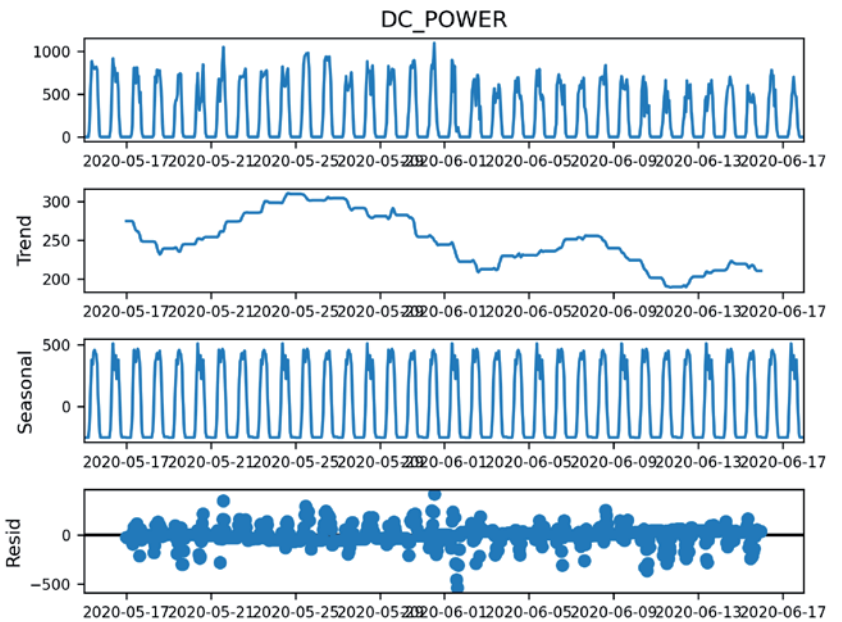


Figure 3.
Seasonal decomposition of DC power.

3.2 Evaluation metrics

Evaluate the performance of the ensemble meta-learning model on the testing data. The performance metrics used include MSE, MAE, the R^2, and run-time, which are mathematically presented below.

$$MSE = \frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2 \tag{8}$$

$$MAE = \frac{1}{n}\sum_{j=1}^{n}\left|y_j - \hat{y}_j\right| \tag{9}$$

$$R^2 = 1 - \frac{\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2}{\sum_{j=1}^{n}\left(y_j - \overline{y}\right)^2} \tag{10}$$

where y_j is the actual value, $\hat{y}_j$ is the predicted value, and $\overline{y}$ is the mean of the actual values.

4. Results and discussion

In this work, we compared the effectiveness of various machine-learning models for solar power forecasting. The base models were KNN, MLP, SVR, GB, and XGB, compared with the proposed ML-SEML model, with LR as the meta-model. The key statistic utilized for comparison was the MSE; in **Figure 4**, we can see that the ML-SEML model has the lowest MSE compared to all other base models, while KNN has the worst MSE. To further examine the performance of our models, we created box plots of the predictions provided by each model

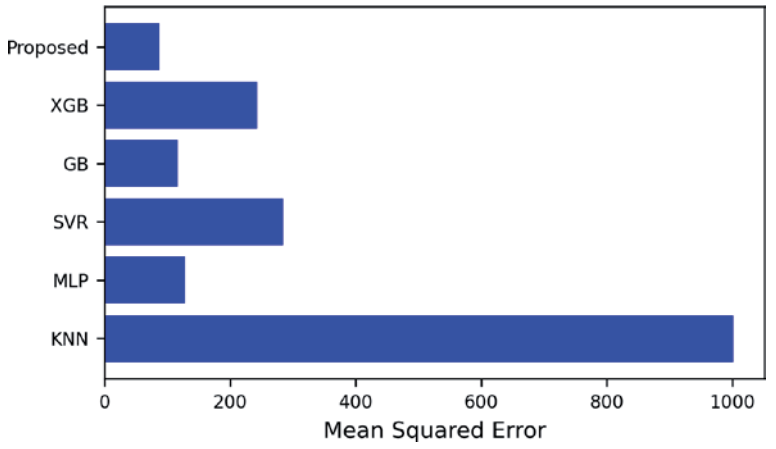


Figure 4.
Comparison of proposed ML-SEML model vs. base models.

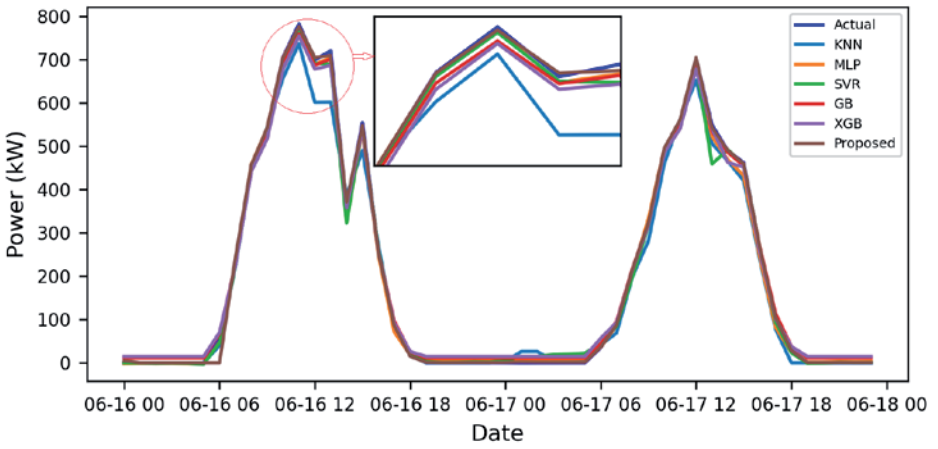


Figure 5.
Solar power forecasting: actual vs. predicted.

and compared them to the actual power values. **Figure 5** depicts a time series of actual and expected solar power levels over 2 days. The graphic shows predictions from both the basic models and the ML-SEML model. The zoomed graph clearly illustrates that ML-SEML model projections are more consistent with real power values, especially during periods of significant fluctuation. This highlights the ML-SEML model's capacity to detect complex patterns and trends in solar power data, which leads to more precise forecasts.

The comparison between our proposed ML-SEML model, base models and the LSTM are summarized in **Table 1**. The proposed model demonstrates superior performance compared to base and LSTM models. With an MSE of 87.25 and MAE of 3.73, the proposed model significantly outperforms LSTM, which has a higher MSE of 265.84 and MAE of 10.97. Additionally, ML-SEML achieves a near-perfect R^2 of 0.999, showcasing its robustness in predictive accuracy, while LSTM lags with an R^2 of 0.995. Furthermore, ML-SEML's runtime of 0.01559 mins is three times faster than LSTM's 0.0483 mins, making it computationally more efficient, thus highlighting its overall dominance.

Models	**Performance Indices**			
	MSE	**MAE**	**R^2**	**Run Time**
KNN	1001.61	18.79	0.985	0.0003
MLP	128.14	7.53	0.998	0.0096
SVR	284.53	8.97	0.995	0.0011
GB	116.73	9.95	0.998	0.00011
XGB	243.02	13.87	0.996	0.0003
LSTM	265.84	10.97	0.995	0.0483
ML-SEML	87.25	3.73	0.999	0.01559

Table 1.
The comparison between proposed ML-SEML, base models, and LSTM.

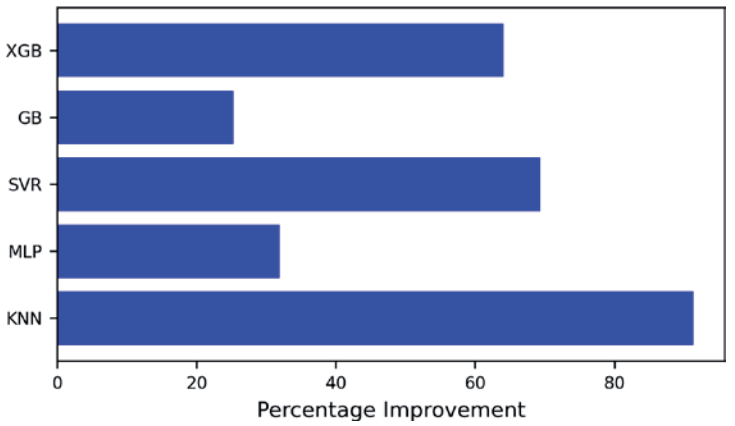


Figure 6.
Percentage improvement of proposed ML-SEML model over base models.

In the final analysis, we determined the MSE-based percentage improvement of the ML-SEML model over each of the base models. The improvements are shown in a bar chart in **Figure 6**. The percentage improvement is computed using the following formula:

$$\text{Improvement} = \left(\frac{MSE_{\text{base}} - MSE_{\text{meta}}}{MSE_{\text{base}}} \right) \times 100 \tag{11}$$

The findings further demonstrate the ML-SEML model's usefulness for SPF by showing that it significantly improves upon base models overall. To sum up, the findings show that our proposed stacked ensemble meta-learning (ML-SEML) model performs much better on all performance measures and routinely beats conventional base models in terms of MSE, MAE, and R^2. It also produces more accurate and dependable predictions.

While the proposed ML-SEML model was evaluated using a specific solar dataset, we acknowledge the need to assess its robustness across various seasonal and geographical contexts rigorously. Variations in solar irradiance, which can arise from factors such as local climate conditions, topographical features, and seasonal fluctuations, can significantly affect the accuracy of solar forecasting. To address this, future research will validate the model using datasets from regions with differing climates, including tropical, temperate, and arid environments. Additionally, we intend to incorporate data that reflects diverse terrain types, such as urban areas, mountainous regions, and open plains, to ensure that our model achieves high generalization and real-world applicability across multiple scenarios. This comprehensive approach aims to enhance the model's predictive capabilities and reliability in practical applications within the solar energy sector.

5. Limitations and practical considerations

The proposed ML-SEML model performs well in short-term solar power forecasting; nonetheless, it is crucial to recognize some limitations. One noticeable limitation

is the model's sensitivity to hyperparameter adjustment, which can considerably impact performance. While we used cross-validation to verify the model's resilience, using hyperparameter optimization approaches like grid search or random search might improve the model's efficiency and accuracy. Furthermore, there is a risk of overfitting, especially when dealing with correlated features in training data. To overcome this issue, we focused on data pretreatment approaches and used cross-validation to reduce overfitting. However, future studies might look at feature selection or regularization approaches to reduce overfitting, particularly when applying the model to larger and more complex datasets.

The proposed ML-SEML model was implemented using Python 3.8. Scikit-learn is used for machine learning methods (KNN, MLP, SVR, Gradient Boosting, and XGBoost), NumPy and Pandas for data processing, and Matplotlib and Seaborn for data visualization. The TensorFlow and Keras libraries (tensorflow.keras.models. Sequential and tensorflow.keras.layers.LSTM, Dense) were used to compare LSTM-based models. Model training and assessment were performed on a conventional workstation without GPU acceleration. The stacking ensemble was built with Scikit-learn's StackingRegressor framework. To guarantee repeatability, random seeds were set throughout training. The workflow consisted of data preprocessing (scaling and normalization), base model training, prediction generation from base models, and final prediction using the meta-learner.

Real-world implementation in grid operations requires careful consideration of a number of realistic issues. First, latency is a crucial consideration. The suggested ML-SEML framework offers good forecasting accuracy, but there may be a little computational burden due to its multi-model design. However, the delay stays within reasonable bounds for integration into energy management systems (EMS) because the model works on short-term forecasting horizons (e.g., 15 minutes to 1 hour ahead). Integrating with the current grid infrastructure presents another difficulty. A real-time data pipeline that can process high-frequency solar irradiance and power measurements must incorporate the forecasting model. Additional data formatting or middleware development may be necessary to ensure compatibility with Distributed Energy Resource Management Systems (DERMS) or Supervisory Control and Data Acquisition (SCADA) systems.

In this study, we employed fundamental feature engineering techniques, such as normalization and the construction of lag features, to effectively capture the temporal dependencies inherent in solar irradiance patterns. Normalization was utilized to scale the data, ensuring that all features contribute equally to the learning process, while lag feature creation helped in encoding past values and trends, which are critical for predicting future irradiance levels. These strategies aimed not only to enhance the predictive accuracy of our models but also to minimize noise that can obscure meaningful patterns in the data. Although we did not incorporate advanced dimensionality reduction techniques such as Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and autoencoders in the current analysis, we recognize their significant potential for future research. These techniques can play a vital role in managing high-dimensional sensor data by reducing redundancy and computational overhead, allowing for more efficient processing and potentially improved model performance. By applying these methods, future studies could uncover deeper insights and facilitate a more streamlined analysis of complex datasets related to solar energy.

Future work will concentrate on overcoming these problems and refining the ML-SEML framework for smooth deployment in operational contexts.

6. Conclusion

In this research, we proposed a machine learning-driven stacked ensemble meta-learning model for solar power forecasting, using KNN, MLP, SVR, GB, and XGB as base models and LR as the meta-model. The ML-SEML model outperformed individual base models and the LSTM model regarding MSE, MAE, R^2, and runtime. The results demonstrated the ML-SEML model's capacity to detect complex patterns and temporal dynamics in solar power data, resulting in more accurate and dependable forecasts than standard models. The box plots and time series graphs demonstrated the ML-SEML model's ability to predict solar power with low variation and greater faithfulness to absolute power levels. Furthermore, the ML-SEML model achieved significant percentage gains over the base models and the LSTM, demonstrating its usefulness in improving solar power forecasting accuracy. The proposed model can predict any of the variables shown in **Figure 2**. Future research should incorporate real-time meteorological data for improved accuracy and develop hybrid models that combine the ML-SEML model with RNNs or LSTMs to ensure scalability and resilience under different conditions.

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Conflict of interest

The authors declare no conflict of interest.

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